

Kshitiz Kumar

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Summary

Software Development Engineer aspirant with hands-on experience in building AI-driven applications, including Retrieval-Augmented Generation (RAG) systems, LLM integrations, and intelligent automation pipelines. Proficient in developing scalable full-stack systems using React.js, Node.js, Express.js, and SQL/NoSQL databases, with a strong foundation in Data Structures and Algorithms, OOP, RESTful APIs, and data-driven system design.

Technical Skills

Languages: C, C++, Java, Python, JavaScript/TypeScript

Frontend: React.js, Tailwind CSS, Bootstrap, HTML5, CSS3

Backend & APIs: Node.js, Fast API, Express.js, RESTful APIs

Databases: MongoDB, MySQL, PostgreSQL (schema design, joins)

Deployment: Vercel, Railway, Render

Tools & Platforms: Git, GitHub, VS Code, Neon, Postman

Projects

Internshala Auto Apply (Automation & Testing Tool) *Node.js, Playwright, Gemini API, JavaScript*

- Engineered a scalable **browser automation system** using Playwright to automate end-to-end internship applications for **50+ listings/session**, simulating real user workflows (authentication, search, form submission) with **95%+ reliability**.
- Integrated **LLM-based response generation (Gemini API)** to dynamically generate answers for **10+ application questions per listing**, reducing manual effort by **80%+** while maintaining semantic relevance.
- Built an **automated outreach pipeline** for HR communication by performing **job description parsing and resume analysis**, generating personalized emails in **5–15 seconds** per application.
- Implemented **keyword extraction and resume optimization pipeline** aligning resumes with **20+ JD-specific keywords**, improving ATS compatibility and ensuring **data consistency and integrity**.

EduPilot – AI Course Assistant *Python, Streamlit, Playwright, FAISS, Hugging Face, Groq API, Llama3*

- Built an **AI-powered RAG system** that answers queries from any educational program URL, eliminating manual search across complex college websites.
- Developed an automated **web crawling pipeline using Playwright** to scrape and aggregate multi-page data (curriculum, fees, admissions) with dynamic sub-page discovery.
- Designed a **semantic retrieval system** using **all-MiniLM-L6-v2 embeddings** and **FAISS indexing**, retrieving **top-8 relevant chunks** for accurate context grounding.
- Integrated **Llama3 via Groq API** to generate **zero-hallucination responses** with source attribution and confidence scoring in a real-time Streamlit interface.

Legal Query Platform (MNLU Collaboration) *React.js, Node.js, Express.js, MongoDB, Tailwind CSS*

- Built a full-stack platform with **10+ pages** enabling users to submit legal queries and receive responses from law students.
- Developed an admin panel handling **user management, query tracking, content updates, and resolution workflow**.
- Coordinated development within a **6-member cross-college team**, performing requirement analysis and task distribution.

Student Placement Prediction System *Python, Random Forest, Pandas, React.js, Node.js*

- Trained a **Random Forest classifier** on **3000+ student records**, achieving **88.55% prediction accuracy**.
- Performed feature engineering on **7+ academic and skill-based attributes** to improve model performance.
- Deployed the system using a **React frontend and Node.js backend** for real-time predictions.

Education

B.Tech in Computer Science and Engineering, Indian Institute of Information Technology, Nagpur Aug 2023 – Aug 2027

CGPA: **7.73 / 10**

Senior Secondary (Class XII), Gautam Public Sr. Sec. School, CBSE

Jul 2020 – Jul 2022

Score: **89.6%**

Achievements

- Solved **400+** Data Structures and Algorithms problems on **LeetCode**.
- Achieved a peak **CodeChef rating of 1473**.